



National
Qualifications
2024

X857/76/11

Physics
Paper 2 — Relationships sheet

THURSDAY, 25 APRIL

10:15 AM – 12:30 PM



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Relationships required for Physics Higher

$$d = \bar{v}t$$

$$s = \bar{v}t$$

$$v = u + at$$

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

$$s = \frac{1}{2}(u + v)t$$

$$F = ma$$

$$W = mg$$

$$E_w = Fd, \text{ or } W = Fd$$

$$E_p = mgh$$

$$E_k = \frac{1}{2}mv^2$$

$$P = \frac{E}{t}$$

$$p = mv$$

$$Ft = mv - mu$$

$$F = G \frac{m_1 m_2}{r^2}$$

$$t' = \frac{t}{\sqrt{1 - \left(\frac{v}{c}\right)^2}}$$

$$l' = l \sqrt{1 - \left(\frac{v}{c}\right)^2}$$

$$f_o = f_s \left(\frac{v}{v \pm v_s} \right)$$

$$z = \frac{\lambda_{\text{observed}} - \lambda_{\text{rest}}}{\lambda_{\text{rest}}}$$

$$z = \frac{v}{c}$$

$$v = H_0 d$$

$$W = QV$$

$$E = mc^2$$

$$I = \frac{P}{A}$$

$$I = \frac{k}{d^2}$$

$$I_1 d_1^2 = I_2 d_2^2$$

$$E = hf$$

$$E_k = hf - hf_0$$

$$v = f\lambda$$

$$E_2 - E_1 = hf$$

$$d \sin \theta = m\lambda$$

$$n = \frac{\sin \theta_1}{\sin \theta_2}$$

$$\frac{\sin \theta_1}{\sin \theta_2} = \frac{\lambda_1}{\lambda_2} = \frac{v_1}{v_2}$$

$$\sin \theta_c = \frac{1}{n}$$

$$V_{rms} = \frac{V_{peak}}{\sqrt{2}}$$

$$I_{rms} = \frac{I_{peak}}{\sqrt{2}}$$

$$T = \frac{1}{f}$$

$$V = IR$$

$$P = IV = I^2 R = \frac{V^2}{R}$$

$$R_T = R_1 + R_2 + \dots$$

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$$

$$V_1 = \left(\frac{R_1}{R_1 + R_2} \right) V_S$$

$$\frac{V_1}{V_2} = \frac{R_1}{R_2}$$

$$E = V + Ir$$

$$C = \frac{Q}{V}$$

$$Q = It$$

$$E = \frac{1}{2}QV = \frac{1}{2}CV^2 = \frac{1}{2} \frac{Q^2}{C}$$

$$\text{path difference} = m\lambda \quad \text{or} \quad \left(m + \frac{1}{2}\right)\lambda \quad \text{where } m = 0, 1, 2, \dots$$

$$\text{random uncertainty} = \frac{\text{max. value} - \text{min. value}}{\text{number of values}}$$

or

$$\Delta R = \frac{R_{\text{max}} - R_{\text{min}}}{n}$$

Additional relationships

Circle

$$\text{circumference} = 2\pi r$$

$$\text{area} = \pi r^2$$

Sphere

$$\text{area} = 4\pi r^2$$

$$\text{volume} = \frac{4}{3}\pi r^3$$

Trigonometry

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

Electron arrangements of elements

Group 1 Group 2

(1)

1 H Hydrogen	4 Be 2,2
3 Li 2,1	12 Mg 2,8,2
Lithium	Beryllium
11 Na 2,8,1	20 Ca 2,8,8,2
Sodium	Calcium
19 K 2,8,8,1	38 Sr 2,8,18,8,2
Potassium	Strontium
37 Rb 2,8,18,8,1	56 Ba 2,8,18,18,8,2
Rubidium	Barium
55 Cs 2,8,18,18,8,1	88 Ra 2,8,18,32,18,8,2
Cesium	Radium

Key

Atomic number
Symbol
Electron arrangement
Name

Transition elements

(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
21 Sc 2,8,9,2	22 Ti 2,8,10,2	23 V 2,8,11,2	24 Cr 2,8,13,1	25 Mn 2,8,13,2	26 Fe 2,8,14,2	27 Co 2,8,15,2	28 Ni 2,8,16,2	29 Cu 2,8,18,1	30 Zn 2,8,18,2
Scandium	Titanium	Vanadium	Chromium	Manganese	Iron	Cobalt	Nickel	Copper	Zinc
39 Y 2,8,18,9,2	40 Zr 2,8,18,10,2	41 Nb 2,8,18,12,1	42 Mo 2,8,18,13,1	43 Tc 2,8,18,13,2	44 Ru 2,8,18,15,1	45 Rh 2,8,18,16,1	46 Pd 2,8,18,18,0	47 Ag 2,8,18,18,1	48 Cd 2,8,18,18,2
Yttrium	Zirconium	Niobium	Molybdenum	Technetium	Ruthenium	Rhodium	Palladium	Silver	Cadmium
57 La 2,8,18,18,9,2	72 Hf 2,8,18,32,10,2	73 Ta 2,8,18,32,11,2	74 W 2,8,18,32,12,2	75 Re 2,8,18,32,13,2	76 Os 2,8,18,32,14,2	77 Ir 2,8,18,32,15,2	78 Pt 2,8,18,32,17,1	79 Au 2,8,18,32,18,1	80 Hg 2,8,18,32,18,2
Lanthanum	Hafnium	Tantalum	Tungsten	Rhenium	Osmium	Iridium	Platinum	Gold	Mercury
89 Ac 2,8,18,32,18,9,2	104 Rf 2,8,18,32,32,10,2	105 Db 2,8,18,32,32,11,2	106 Sg 2,8,18,32,32,12,2	107 Bh 2,8,18,32,32,13,2	108 Hs 2,8,18,32,32,14,2	109 Mt 2,8,18,32,32,15,2	110 Ds 2,8,18,32,32,17,1	111 Rg 2,8,18,32,32,18,1	112 Cn 2,8,18,32,32,18,2
Actinium	Rutherfordium	Dubnium	Seaborgium	Bohrium	Hassium	Meitnerium	Darmstadtium	Roentgenium	Copernicium

5 B 2,3	6 C 2,4	7 N 2,5	8 O 2,6	9 F 2,7	10 Ne 2,8
Boron	Carbon	Nitrogen	Oxygen	Fluorine	Neon
13 Al 2,8,3	14 Si 2,8,4	15 P 2,8,5	16 S 2,8,6	17 Cl 2,8,7	18 Ar 2,8,8
Aluminium	Silicon	Phosphorus	Sulfur	Chlorine	Argon
31 Ga 2,8,18,3	32 Ge 2,8,18,4	33 As 2,8,18,5	34 Se 2,8,18,6	35 Br 2,8,18,7	36 Kr 2,8,18,8
Gallium	Germanium	Arsenic	Selenium	Bromine	Krypton
49 In 2,8,18,18,3	50 Sn 2,8,18,18,4	51 Sb 2,8,18,18,5	52 Te 2,8,18,18,6	53 I 2,8,18,18,7	54 Xe 2,8,18,18,8
Indium	Tin	Antimony	Tellurium	Iodine	Xenon
81 Tl 2,8,18,32,18,3	82 Pb 2,8,18,32,18,4	83 Bi 2,8,18,32,18,5	84 Po 2,8,18,32,18,6	85 At 2,8,18,32,18,7	86 Rn 2,8,18,32,18,8
Thallium	Lead	Bismuth	Polonium	Astatine	Radon

Group 3 Group 4 Group 5 Group 6 Group 7 Group 0

(18)

57 La 2,8,18,18,9,2	58 Ce 2,8,18,20,8,2	59 Pr 2,8,18,21,8,2	60 Nd 2,8,18,22,8,2	61 Pm 2,8,18,23,8,2	62 Sm 2,8,18,24,8,2	63 Eu 2,8,18,25,8,2	64 Gd 2,8,18,25,9,2	65 Tb 2,8,18,27,8,2	66 Dy 2,8,18,28,8,2	67 Ho 2,8,18,29,8,2	68 Er 2,8,18,30,8,2	69 Tm 2,8,18,31,8,2	70 Yb 2,8,18,32,8,2	71 Lu 2,8,18,32,9,2
Lanthanum	Cerium	Praseodymium	Neodymium	Promethium	Samarium	Europium	Gadolinium	Terbium	Dysprosium	Holmium	Erbium	Thulium	Ytterbium	Lutetium
89 Ac 2,8,18,32,18,9,2	90 Th 2,8,18,32,18,10,2	91 Pa 2,8,18,32,20,9,2	92 U 2,8,18,32,21,9,2	93 Np 2,8,18,32,22,9,2	94 Pu 2,8,18,32,24,8,2	95 Am 2,8,18,32,25,8,2	96 Cm 2,8,18,32,25,9,2	97 Bk 2,8,18,32,27,8,2	98 Cf 2,8,18,32,28,8,2	99 Es 2,8,18,32,29,8,2	100 Fm 2,8,18,32,30,8,2	101 Md 2,8,18,32,31,8,2	102 No 2,8,18,32,32,8,2	103 Lr 2,8,18,32,32,9,2
Actinium	Thorium	Protactinium	Uranium	Neptunium	Plutonium	Americium	Curium	Berkelium	Californium	Einsteinium	Fermium	Mendelevium	Nobelium	Lawrencium

Lanthanides

Actinides